C241-PS369 | CAPSTONE PRODUCT

MOODIFY

MENTAL HEALTH TRACKING AND ASSISTANCE APPLICATION



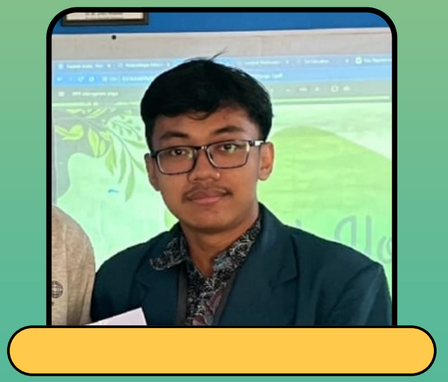
Meet The Team



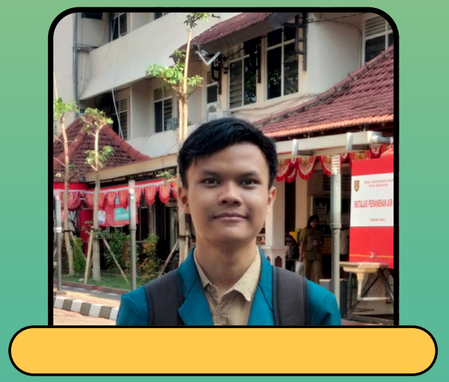
M Javier Rasyadi
Mobile Development



Jauza Aryazuhdi A
Mobile Development



Fitriawan Arya Nugraha
Cloud Computing



M. Bintang Prayoga U
Cloud Computing



Asnawi Alamsyah
Machine Learning



Fachry Syifa A
Machine Learning



Sultan Fahd MBY
Machine Learning

Overview

1. Background Story
2. Reason
3. Existing Result
4. Implementation
5. Result
6. Documentation
7. Local Deployment Plan



moodify.dev

Scan to explore our product!

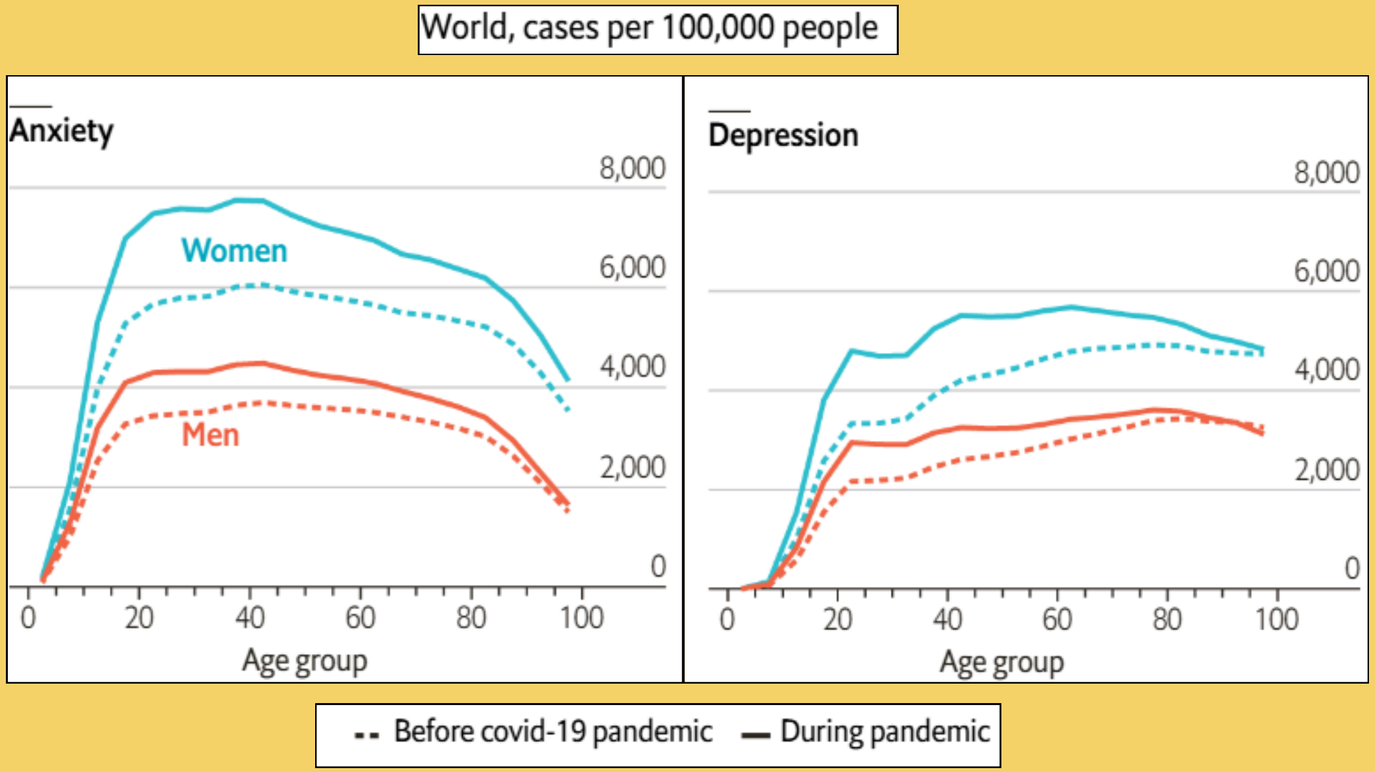
BACKGROUND STUDY

1



Impact of the COVID-19 Pandemic on Global Mental Health

The COVID-19 pandemic has significantly increased global rates of major depressive disorder by 27.6%, resulting in 53.2 million additional cases, and anxiety disorders by 25.6%, leading to 76.2 million more cases. These figures highlight a critical need for increased mental health support worldwide.



Source: “Global prevalence and burden of depressive and anxiety disorders in 204 countries and territories in 2020 due to the covid-19 pandemic”, by D.F. Santomauro et al., The Lancet, 2021

High prevalence of mental health issues in Indonesia



Image source : indonesiabaik.id

6.1% of the population aged >15 years suffer from depression, and 6.7% of households have members with mental disorders such as schizophrenia or psychosis (Riskesdas, 2018).

2

REASON



Why Moodify?

Our Inspiration

- Our concern for Gen Z's mental health challenges.
- Recognizing mental health as essential to overall well-being.

Our Goals

- Develop an app to identify and understand users' moods through their journals.
- Provide emotional management tools and coping strategies.



Problems We Aim to Tackle

Personalized Mental Health Support

- Adolescents often lack tailored mental health resources.
- One-size-fits-all approaches are ineffective.

Effective Mood Tracking and Coping Mechanisms

- Need for tools to track moods and suggest appropriate coping mechanisms.



EXISTING RESULT

3



NATURAL LANGUAGE PROCESSING APPLIED TO MENTAL ILLNESS DETECTION: A NARRATIVE REVIEW

by Tianlin Zhang, Annika M. Schoene, Shaoxiong Ji & Sophia Ananiadou

The research investigates the use of natural language processing (NLP) for mental illness detection. The findings show that NLP methods are beneficial for improving mental healthcare and early diagnosis.



MOBILE APPLICATIONS FOR MENTAL HEALTH SELF-CARE: A SCOPING REVIEW

Arfan Ahmed et. al (2021)

The research discusses the growing prevalence of mental illness and the limitations of traditional treatment. Mobile apps offer a promising solution by providing assessment tools, tracking symptoms, and delivering interventions.



IDENTIFY MISSING ELEMENTS FROM THE PRIOR RESEARCH

An element that does not yet exist in the application that has been created is a place for users to tell what the user feels in a journal that can detect the user's mood through the user's writing. Furthermore, the results of writing the journal are recommendations for coping tools.



IMPLEMENTATION

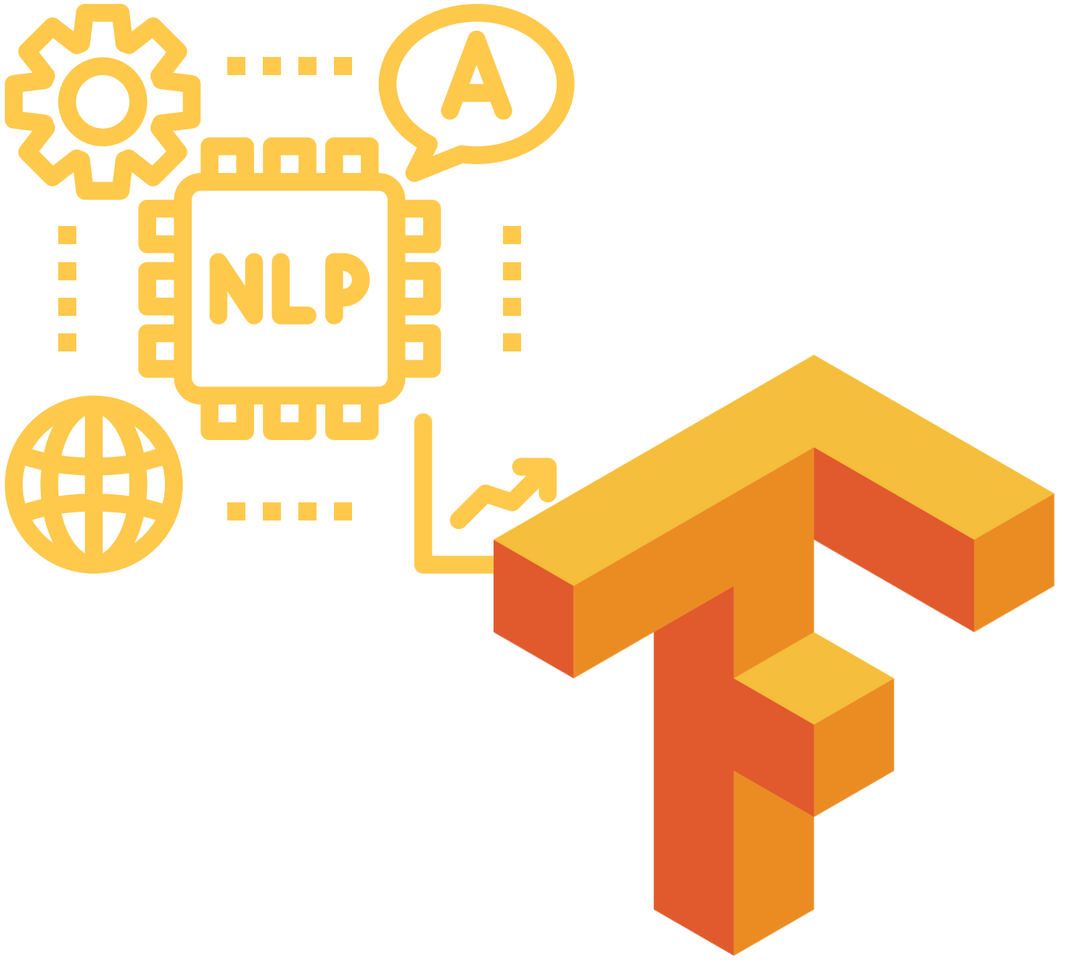
4



Moodify aims to enhance mental health awareness by providing innovative mental health tracking and personalized coping tool recommendations application.



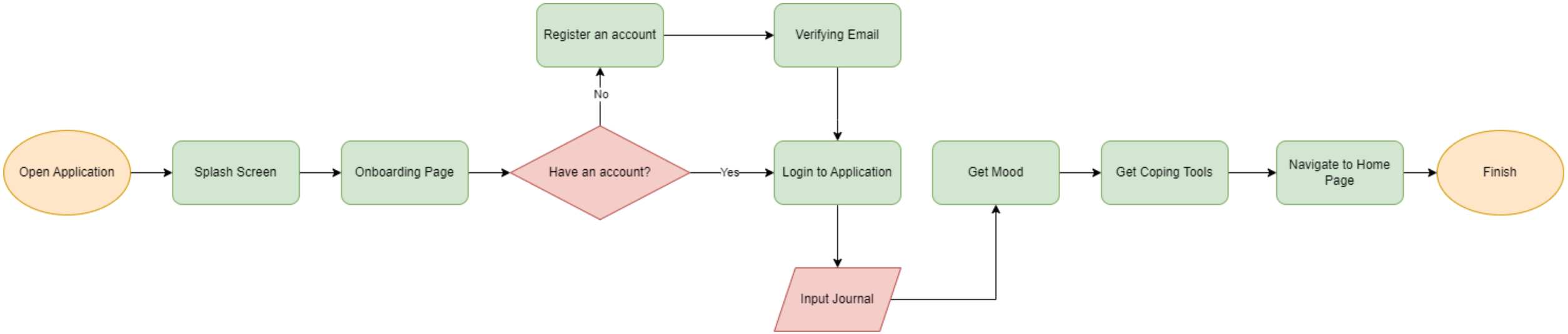
Moodify aspires to accurately detect users' moods from their journal entries by utilizing Natural Language Processing Technology.



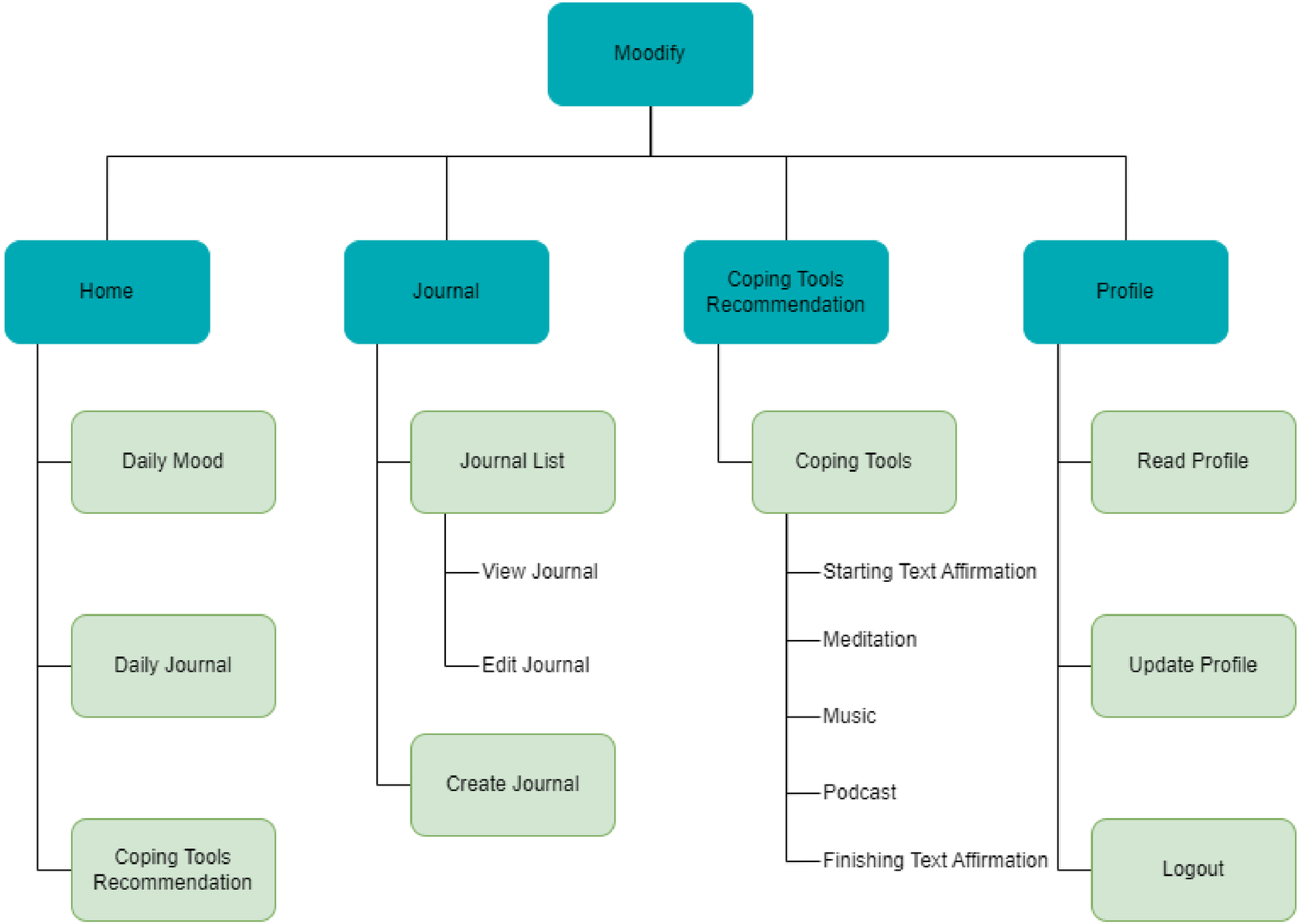
Moodify is committed to delivering reliable and secure services, ensuring users receive effective and personalized mental health support every day.



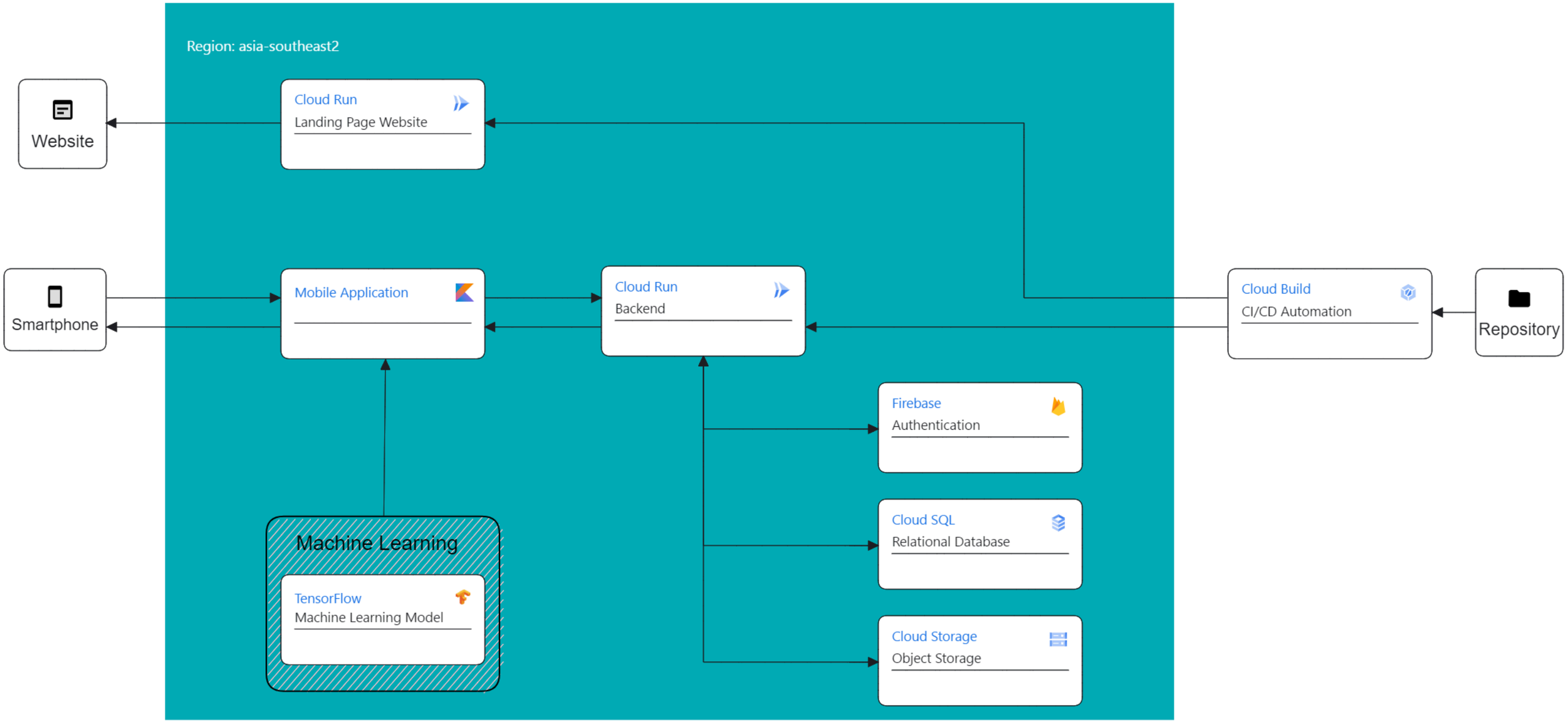
USER FLOWCHART



INFORMATION ARCHITECTURE



ARCHITECTURE DESIGN



5

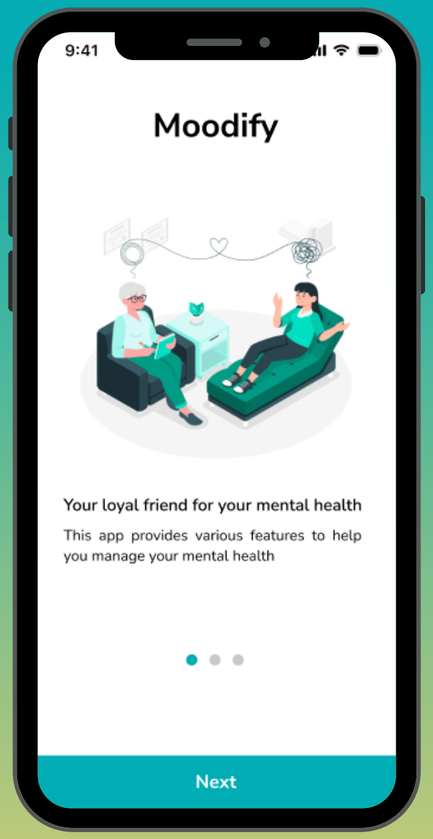
RESULT



Boarding Pages



Splash Screen



1

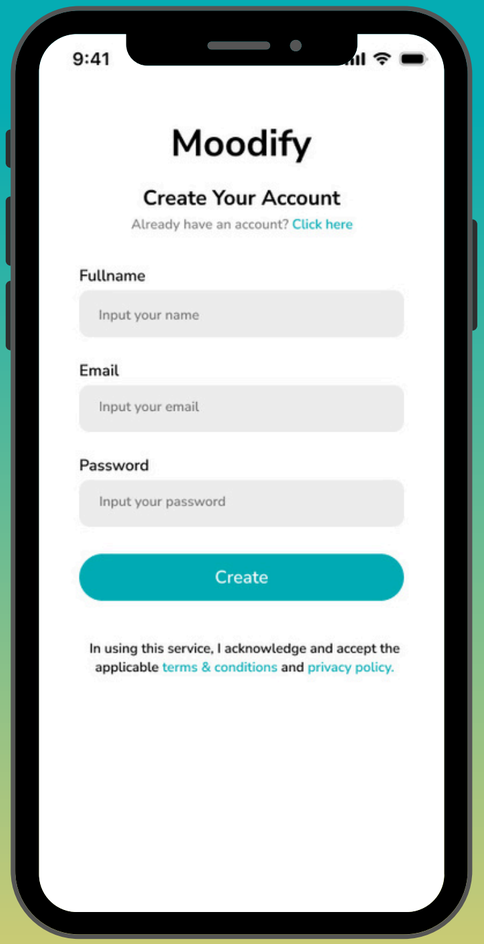


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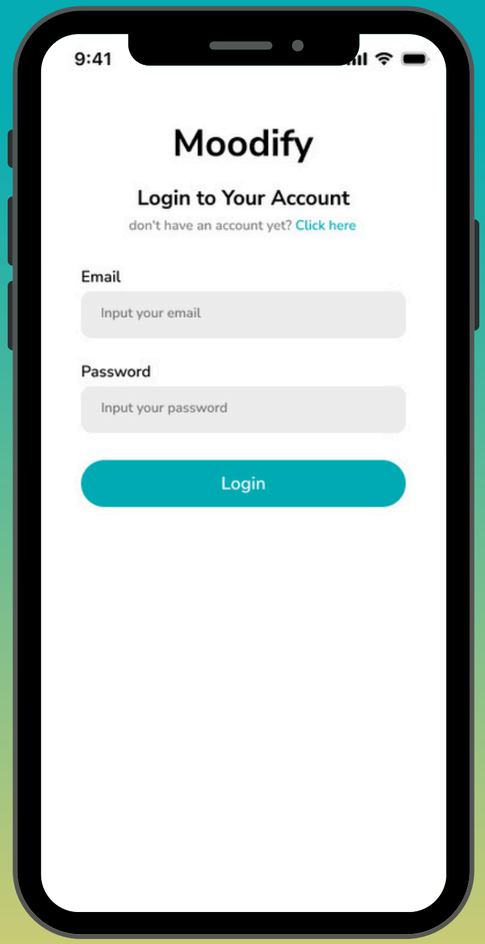


3

Authentication

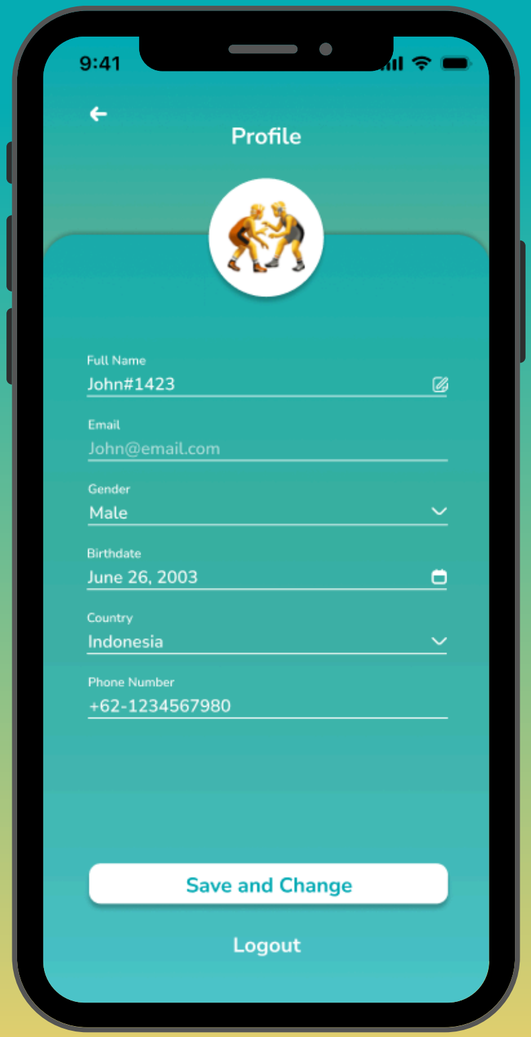


Registration

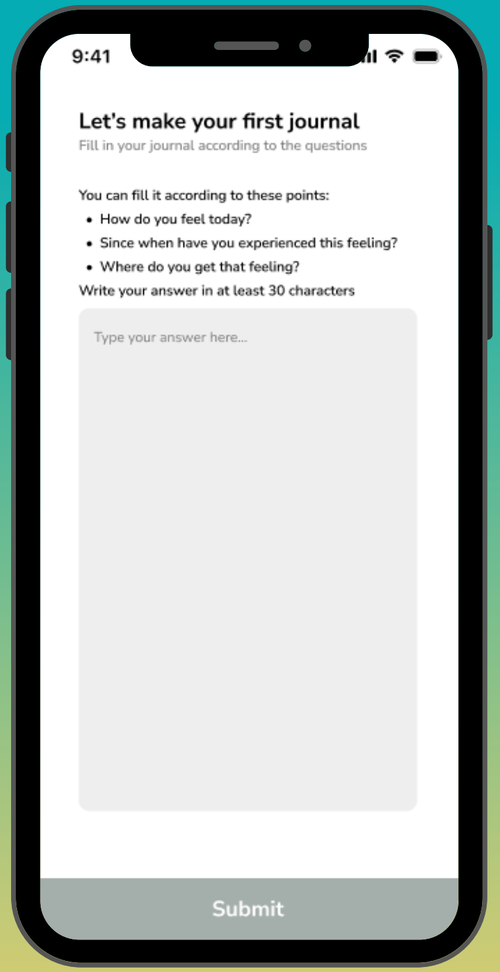


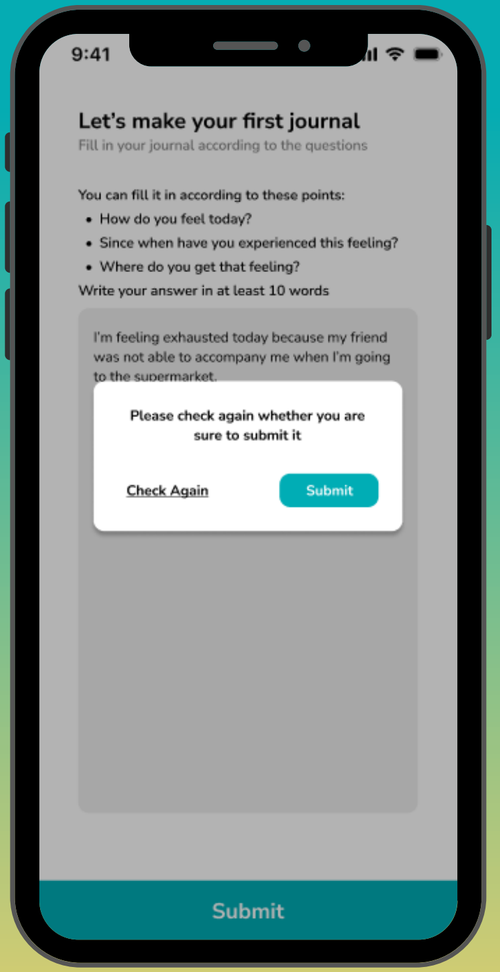
Login

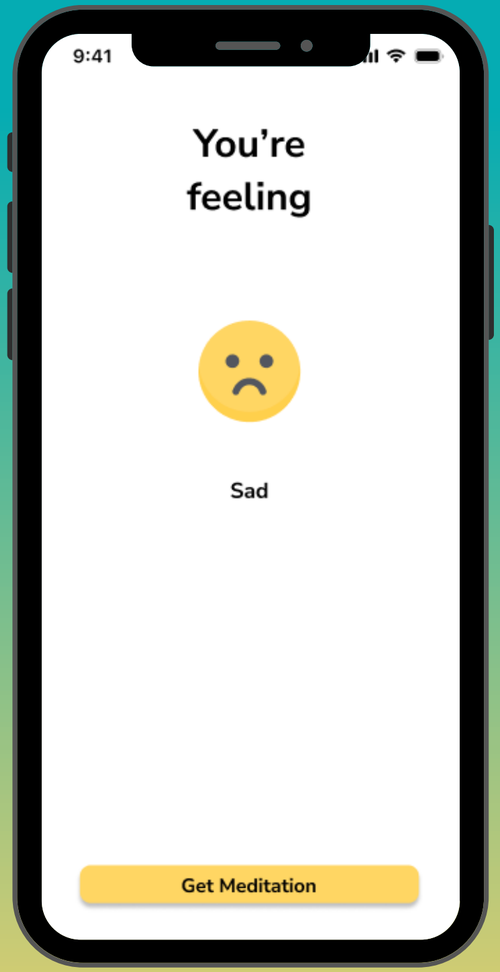
User Profile



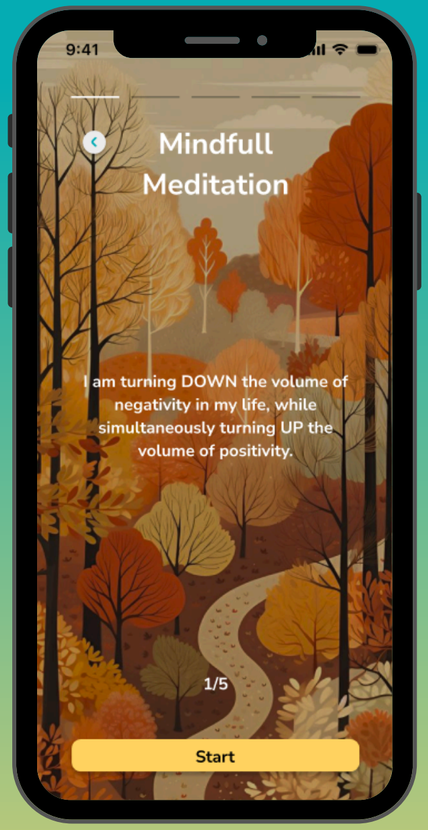
Starter Journal

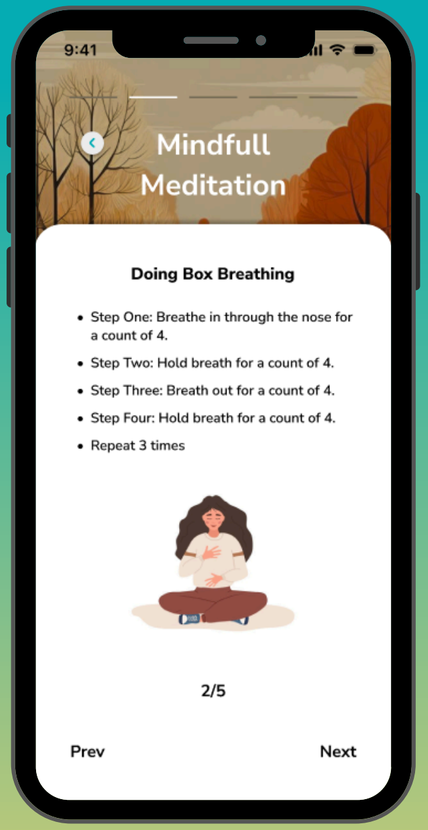


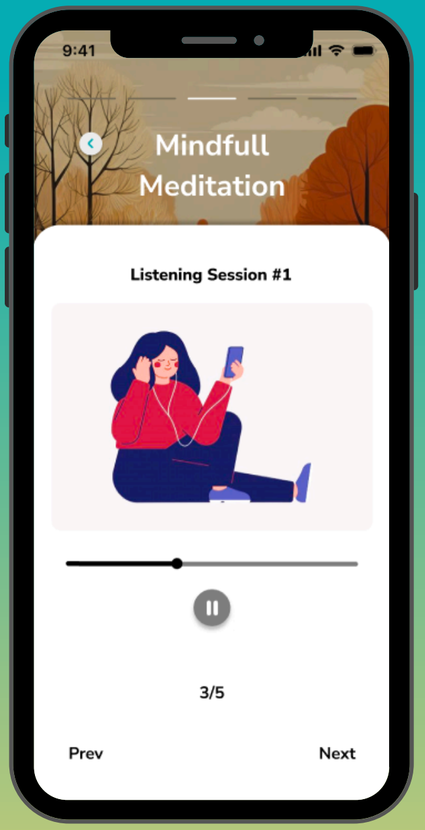


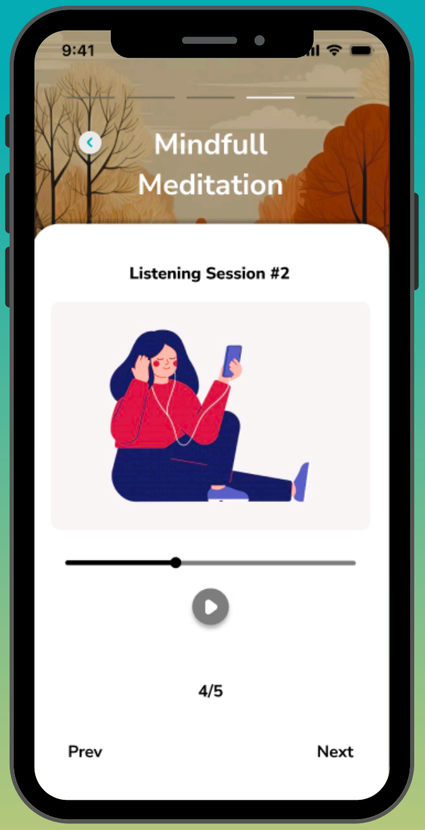


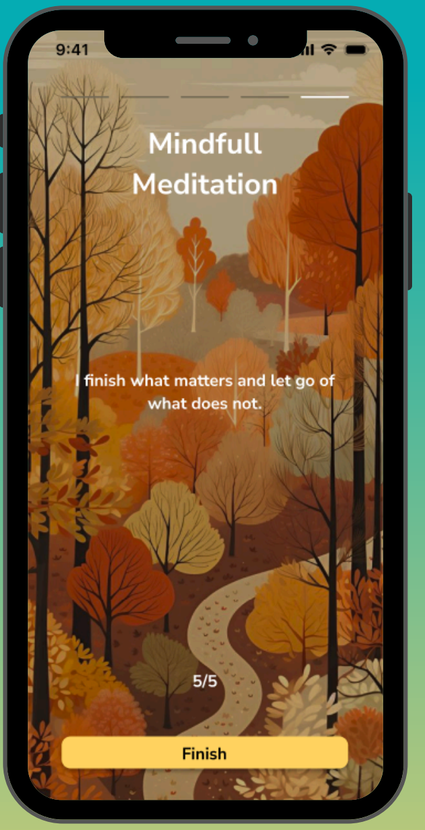
Coping Tools



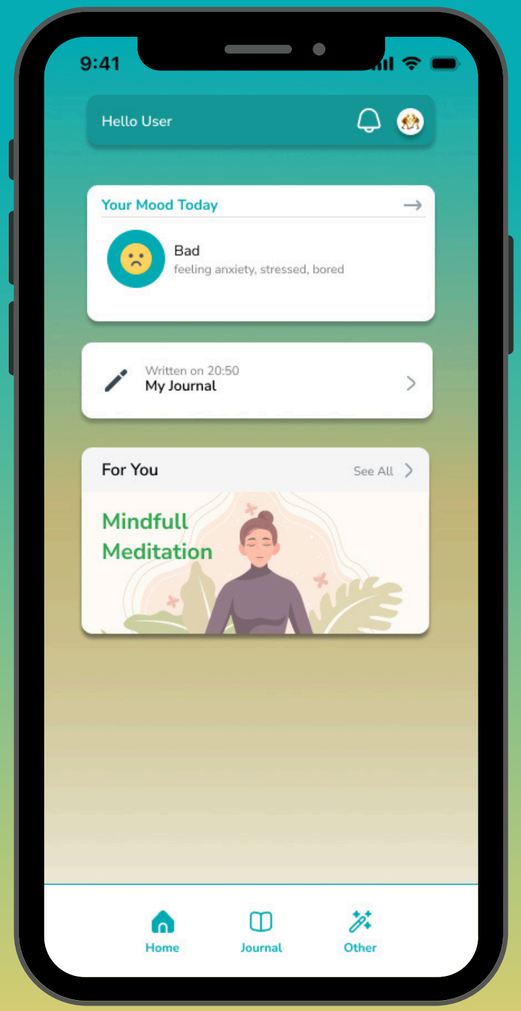




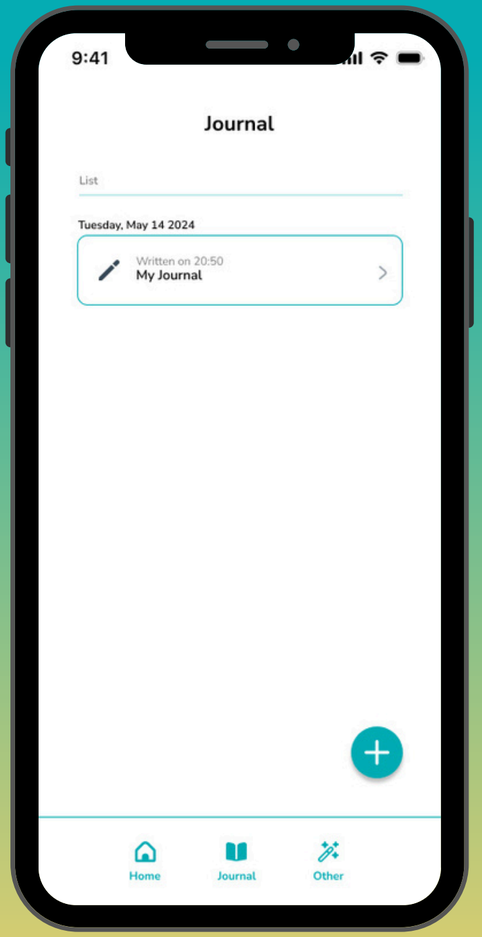


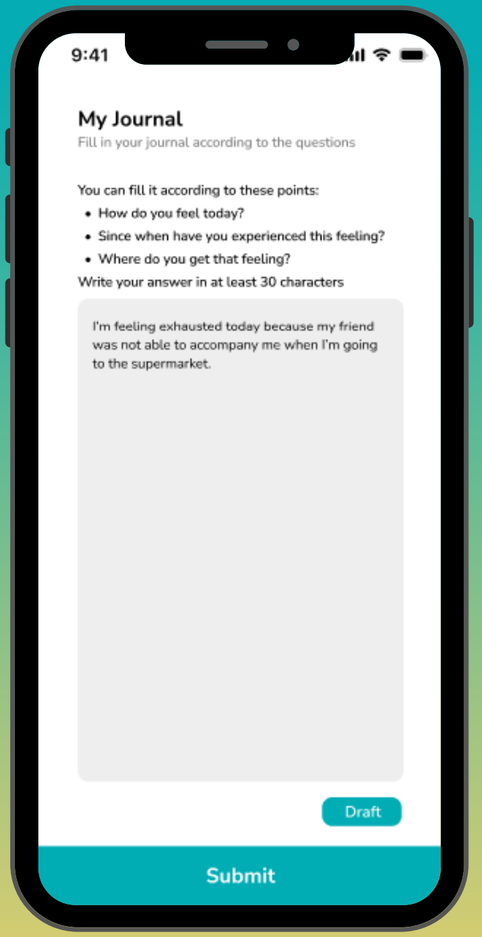


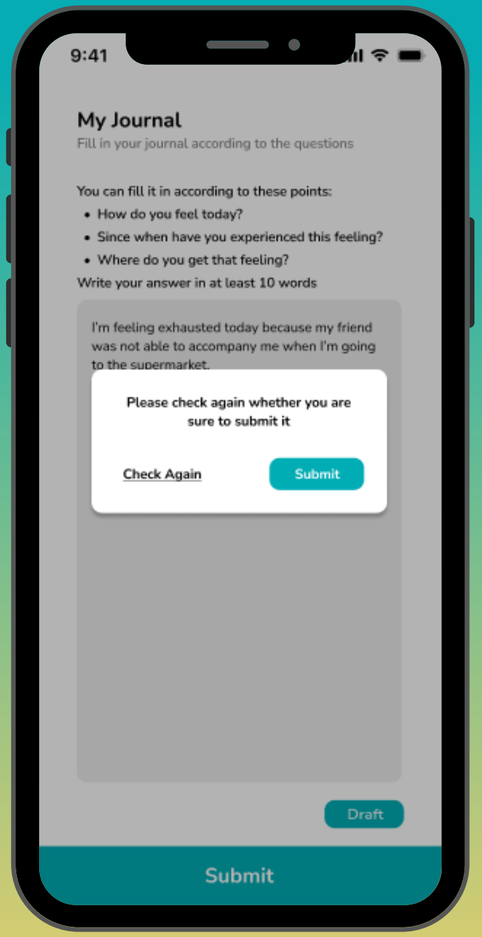
Home Page


9:41
Hello User
Your Mood Today
Bad
feeling anxiety, stressed, bored
Written on 20:50
My Journal
For You
See All
Mindfull
Meditation
Home
Journal
Other

Journal

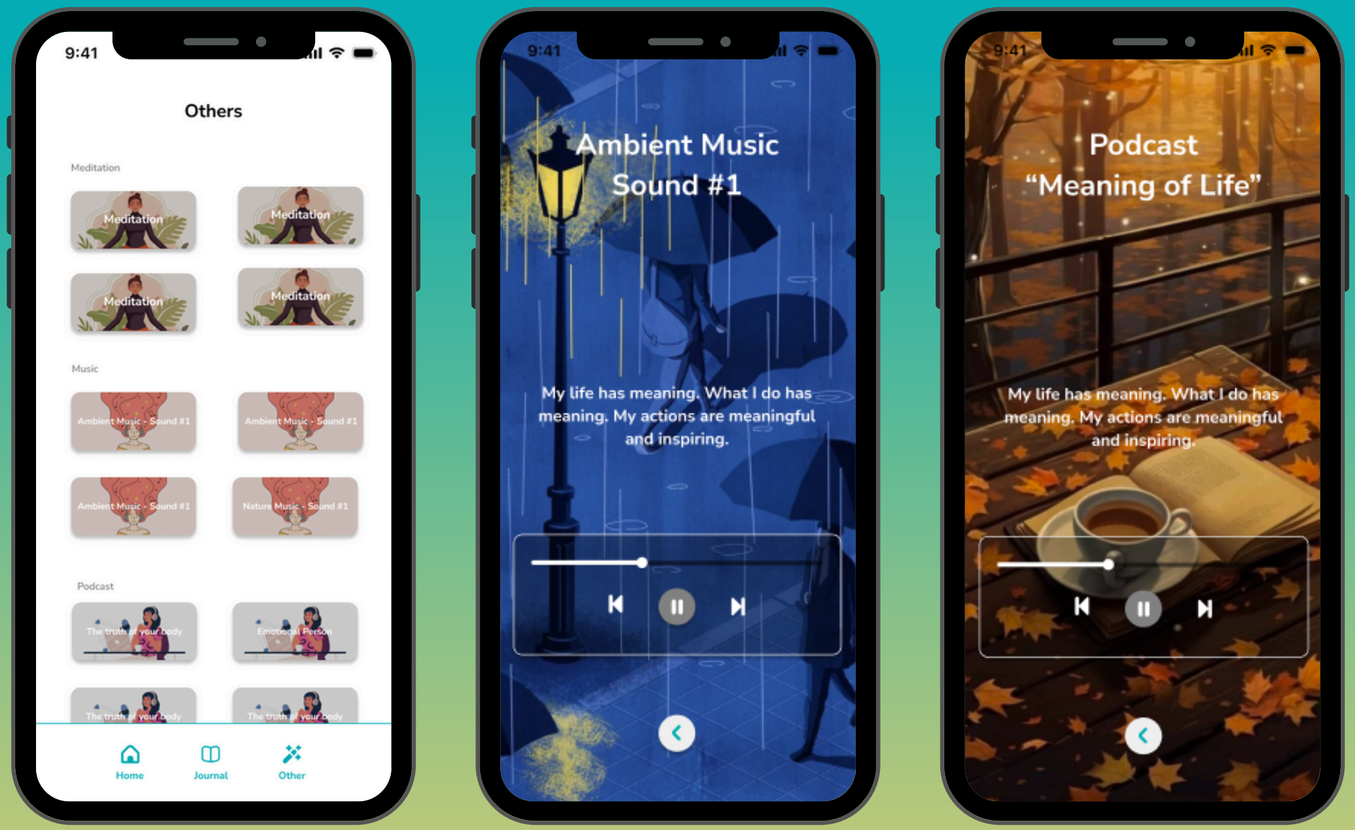

9:41
Journal
List
Tuesday, May 14 2024
Written on 20:50
My Journal
Home
Journal
Other


9:41
My Journal
Fill in your journal according to the questions
You can fill it according to these points:
• How do you feel today?
• Since when have you experienced this feeling?
• Where do you get that feeling?
Write your answer in at least 30 characters
I'm feeling exhausted today because my friend was not able to accompany me when I'm going to the supermarket.
Draft
Submit

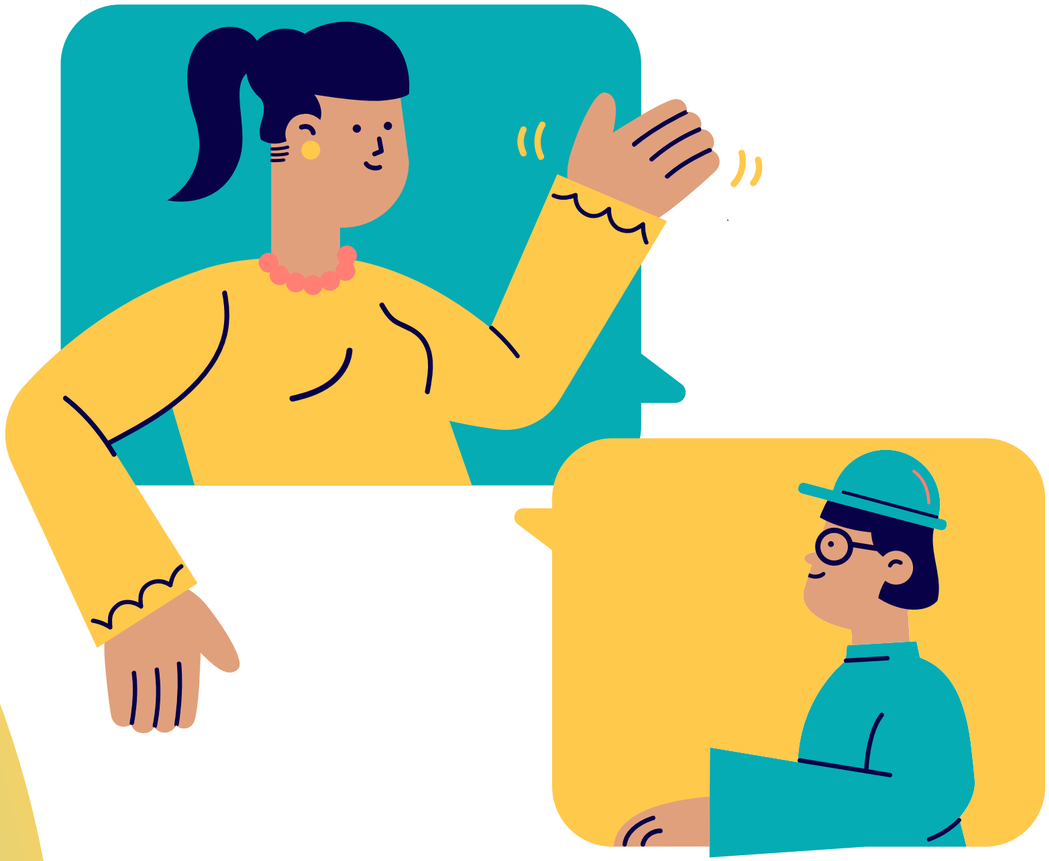

9:41
My Journal
Fill in your journal according to the questions
You can fill it in according to these points:
• How do you feel today?
• Since when have you experienced this feeling?
• Where do you get that feeling?
Write your answer in at least 10 words
I'm feeling exhausted today because my friend was not able to accompany me when I'm going to the supermarket.
Please check again whether you are sure to submit it
Check Again
Submit
Draft
Submit


9:41
LET'S
READ YOUR JOURNAL
Today's Journal
Written on Tuesday, 14 May 2024
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed vestibulum, leo vitae malesuada aliquam, ante ligula mollis ligula, sit amet venenatis leo lectus id ipsum. Suspendisse dictum mauris id mauris sodales, in imperdiet neque vestibulum. Nulla pharetra massa odio, non scelerisque nibh luctus non.
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Save

Other Coping Tools



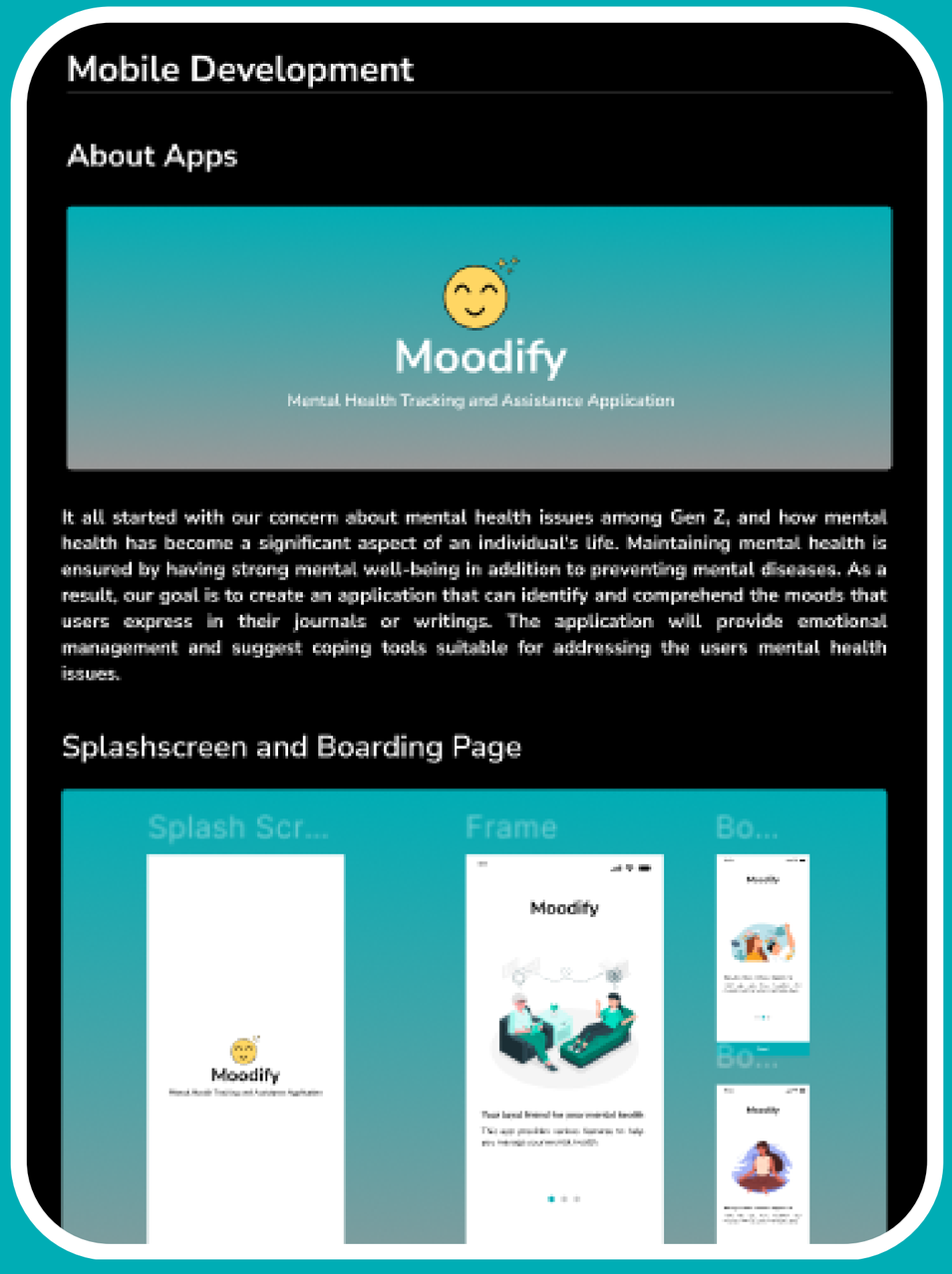
6



DOCUMENTATION

Mobile Development Documentation

https://github.com/awiawii/Moodify/blob/Mobile-Development/README.md



Cloud Computing Documentation

https://github.com/awiawii/Moodify/blob/Cloud-Computing/README.md



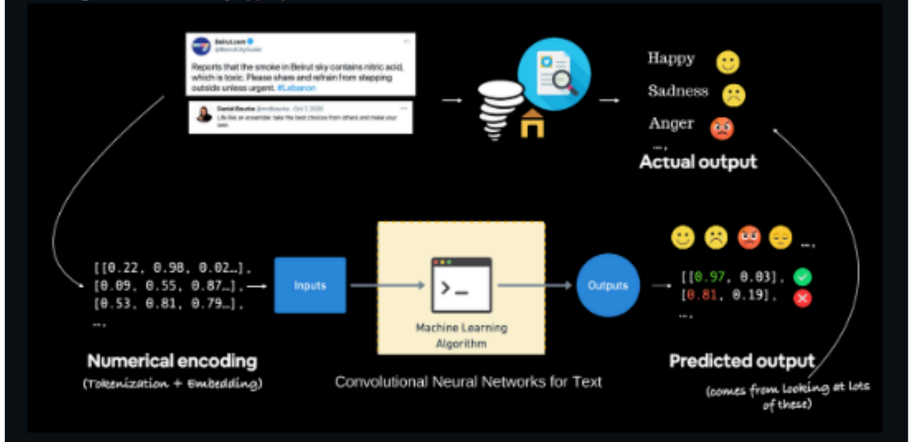
Machine Learning Documentation

https://github.com/awiawii/Moodify/blob/Machine-Learning/README.md

Machine Learning

Natural Language Processing

A Natural Language Processing (NLP) model is a type of artificial intelligence designed to understand, interpret, and generate human language. These models process and analyze large amounts of natural language data to extract meaningful insights and patterns. NLP models play a crucial role in bridging the gap between human communication and computer understanding, enabling machines to comprehend and respond to text or speech in a way that is both meaningful and contextually appropriate.



Dataset

We are using a dataset that is available on Kaggle. Using a dataset available on Kaggle can be an excellent way to implement and experiment with NLP models.

1. Emotions

"Emotions" dataset – a collection of English Twitter messages meticulously annotated with six fundamental emotions: anger, fear, joy, love, sadness, and surprise. This dataset serves as a valuable resource for understanding and analyzing the diverse spectrum of emotions expressed in short-form text on social media.

2. Emotion Detection from Text

The data that we have is having 13 different emotion 40000 records. So it's challenging to build an efficient multiclass classification model. We may need to logically reduce the number of classes here and use some advanced methods to build efficient model.

Emotion in Text

LOCAL DEPLOYMENT PLAN

7



Our Plan

Deployment

Launch and promote **Moodify** across **Indonesia** in phases Finalize development beta launch for feedback and full launch with ongoing promotion

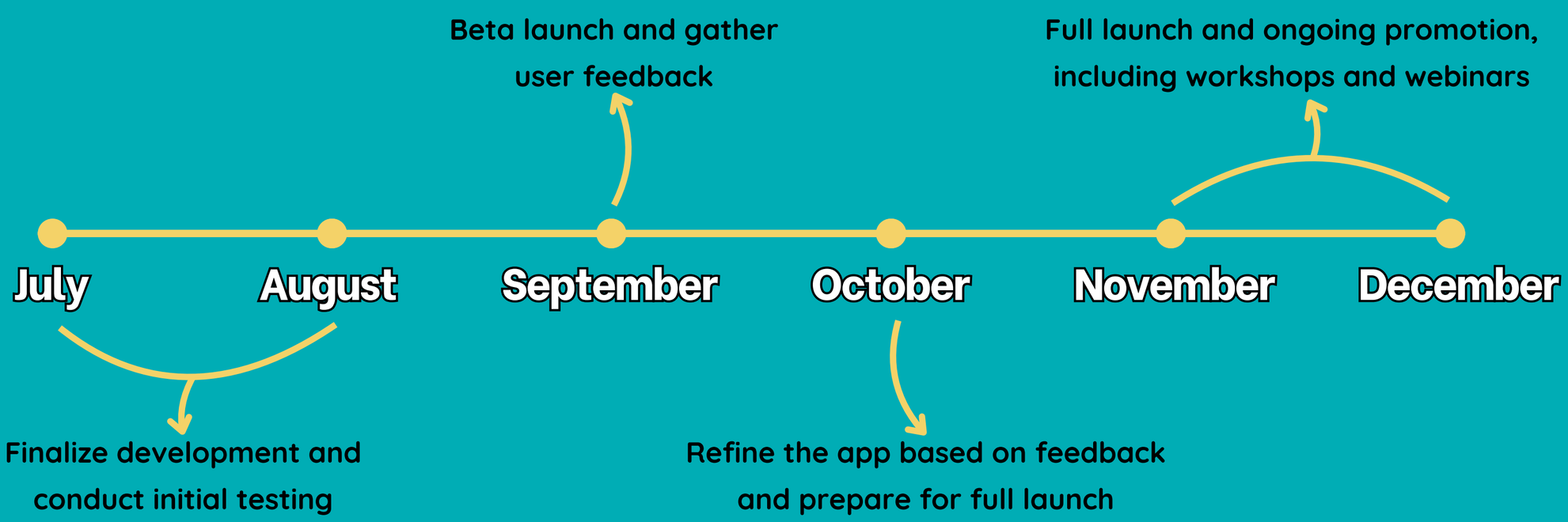
Target Key Regions

Focus on **regions with high mental health issues** using localized campaigns to provide immediate support and achieve significant impact

Collaborate Locally

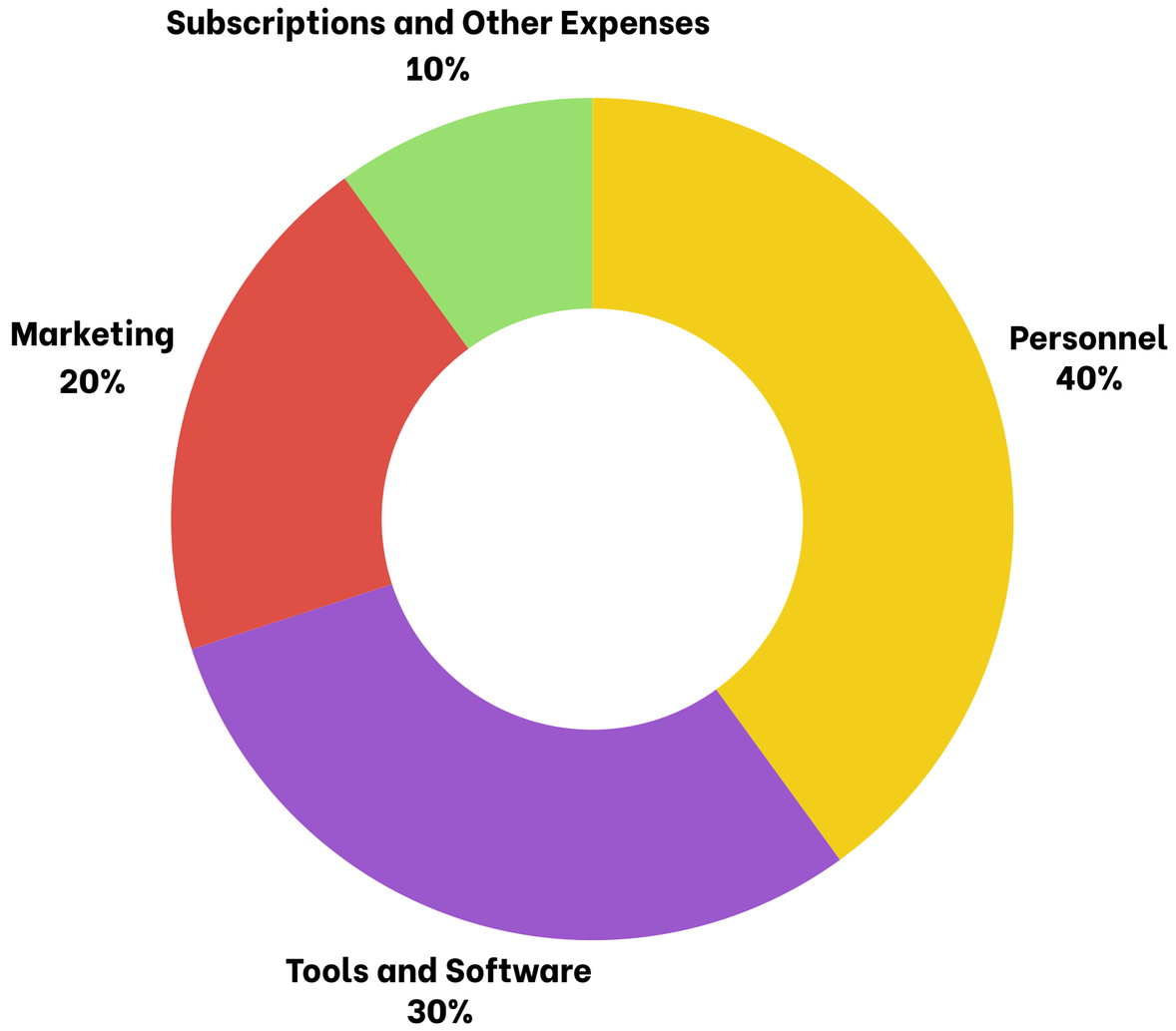
Partner with health organizations and schools to reach adolescents using campaigns and workshops to promote Moodify and support mental health

Project Timeline



Budget Plan $10,000

Personnel	**$4000**
Development team	$2000
Marketing team	$1000
Support staff	$1000
Tools and Software	**$3000**
App development tools	$1500
Analytics software	$1500
Marketing	**$2000**
Online advertising	$1000
Community events and workshops	$500
Promotional materials	$500
Subscriptions and Other Expenses	**$1000**
Hosting and server costs	$500
App store fees	$300
Miscellaneous	$200





Thank you

